Arithmetical Functions

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Throughout these notes,

- m, n, k, d, a, b, c will mean positive integers (unless otherwise specified),
- p will mean a prime number,
- $p_1, p_2, \ldots, p_k$ will mean k pairwise distinct primes,
- $a_1, a_2, \ldots, a_k$ will mean k (not necessarily distinct) positive integers,
- $n = p_1^{a_1} p_2^{a_2} \cdot \ldots \cdot p_k^{a_k}$ will mean the prime factorization of n,
- $\sum_{d|n}$ will mean the sum is taken on all positive divisors d of n,
- $\log n$ will mean the natural log of n,

1 Definitions

Definition 1. A real-valued or complex-valued function defined on the positive integers is called an *arithmetical function*.

Definition 2. The *divisor function* $\sigma_a(n)$ is defined as the sum of the ath powers of the divisors of n, i.e.

$$\sigma_a(n) = \sum_{d|n} d^a.$$

Usually, $\sigma_0(n)$ (the number of the divisors of n) and $\sigma_1(n)$ (the sum of the divisors of n) are denoted by $\tau(n)$ and $\sigma(n)$, respectively, i.e.

$$\tau(n) = \sum_{d|n} 1 \quad \text{and} \quad \sigma(n) = \sum_{d|n} d.$$

Definition 3. The *Euler totient function* $\varphi(n)$ is defined to be the number of positive integers not exceeding n which are relatively prime to n, i.e.

$$\varphi(n) = \sum_{k=1}^{n} \left\lfloor \frac{1}{\gcd(n,k)} \right\rfloor.$$

Definition 4. The *Möbius function* $\mu(n)$ is defined as

$$\mu(n) = \begin{cases} 1 & \text{if } n = 1, \\ (-1)^k & \text{if } n = p_1^{a_1} p_2^{a_2} \cdot \ldots \cdot p_k^{a_k} \text{ and } a_1 = a_2 = \ldots = a_k = 1, \\ 0 & \text{otherwise.} \end{cases}$$

Definition 5. The arithmetical function $I(n)$ given by

$$I(n) = \left\lfloor \frac{1}{n} \right\rfloor = \begin{cases} 1 & \text{if } n = 1, \\ 0 & \text{if } n > 1, \end{cases}$$

is called the *(Dirichlet) identity function*.

Definition 6. The arithmetical function $u(n)$ given by $u(n) = 1$ for all n is called the *unit function*.

Definition 7. The *Liouville function* $\lambda(n)$ is defined as

$$\lambda(n) = \begin{cases} 1 & \text{if } n = 1, \\ (-1)^{a_1 + a_2 + \ldots + a_k} & \text{if } n = p_1^{a_1} p_2^{a_2} \cdot \ldots \cdot p_k^{a_k}. \end{cases}$$

Definition 8. The *von Mangoldt function* $\Lambda(n)$ is defined as

$$\Lambda(n) = \begin{cases} \log p & \text{if } n = p^m \text{ for some prime } p \text{ and some } m \geqslant 1, \\ 0 & \text{otherwise.} \end{cases}$$

Definition 9. An arithmetical function f is said to be *multiplicative* if f is not identically zero and if

$$f(mn) = f(m)f(n) \quad \text{whenever} \quad \gcd(m,n) = 1.$$

Definition 10. An arithmetical function f is said to be *completely multiplicative* if f is not identically zero and if

$$f(mn) = f(m)f(n) \quad \text{for all } \ m, n.$$

Definition 11. The *Dirichlet product* (or *Dirichlet convolution*) of two arithmetical function f and g, denoted by $f * g$, is defined as

$$(f * g)(n) = \sum_{d|n} f(d) g\left(\frac{n}{d}\right).$$

Definition 12. Let f be an arithmetical function with $f(1) \neq 0$. The *Dirichlet inverse* of f, denoted by f^{-1}, is defined as the unique arithmetical function such that

$$f * f^{-1} = f^{-1} * f = I.$$

Definition 13. The *Bell series of an arithmetical function f modulo p*, denoted by f_p, is defined by the formal power series

$$f_p(x) = \sum_{n=0}^{\infty} f(p^n) x^n.$$

Definition 14. The *derivative* of an arithmetical function f, denoted by f', is given by

$$f'(n) = f(n) \log n.$$

2 Theorems

Theorem 1. Let $n = p_1^{a_1} p_2^{a_2} \cdot \ldots \cdot p_k^{a_k}$. Then, for $n \geqslant 1$,

(a) $\sigma_a(n) = \begin{cases} \prod_{i=1}^{k} (a_i + 1) & \text{if } a = 0, \\ \prod_{i=1}^{k} (1 + p_i^a + \ldots + p_i^{a a_i}) & \text{if } a \neq 0, \end{cases}$

(b) $\sum_{d|n} \varphi(d) = n$,

(c) $\varphi(n) = \sum_{d|n} \mu(d) \frac{n}{d}$,

(d) $\sum_{d|n} \mu(d) = I(n)$,

(e) $\varphi(n) = n \prod_{p|n} \left(1 - \frac{1}{p}\right)$,

(f) $\log n = \sum_{d|n} \Lambda(d)$,

(g) $\Lambda(n) = -\sum_{d|n} \mu(d) \log d$.

Theorem 2. For all arithmetical functions f, g, h,

(a) $f * I = I * f = f$

(b) $f * g = g * f$,

(c) $f * (g * h) = (f * g) * h$.

Theorem 3. The Dirichlet inverse of f is given by the recursion formulas

$$f^{-1}(1) = \frac{1}{f(1)}, \quad f^{-1}(n) = -\frac{1}{f(n)} \sum_{\substack{d|n \\ d<n}} f\left(\frac{n}{d}\right) f^{-1}(d) \quad \text{for } n > 1.$$

Theorem 4. The arithmetical functions u and μ are inverses of each other, i.e. $u^{-1}(n) = \mu(n)$ and $\mu^{-1}(n) = u(n)$.

Theorem 5 (Möbius Inversion Formula)**.** For all arithmetical functions f, g,

$$f(n) = \sum_{d|n} g(d) \quad \text{if and only if} \quad g(n) = \sum_{d|n} f(d) \mu\left(\frac{n}{d}\right)$$

Theorem 6. If an arithmetical function f is multiplicative, then $f(1) = 1$.

Theorem 7. Let f be an arithmetical function with $f(1) = 1$. Then

(a) f is multiplicative if and only if

$$f(p_1^{a_1} p_2^{a_2} \cdot \ldots \cdot p_k^{a_k}) = f(p_1^{a_1}) f(p_2^{a_2}) \cdot \ldots \cdot f(p_k^{a_k})$$

for all primes $p_1, p_2, \ldots, p_k$ and all integers $a_1, a_2, \ldots, a_k \geqslant 1$.

(b) if f is multiplicative, then f is completely multiplicative if and only if $f(p^a) = (f(p))^a$. for all primes p and all integers $a \geqslant 1$.

Theorem 8. Let f be a multiplicative arithmetical function. Then

(a) f is completely multiplicative if and only if $f^{-1}(n) = \mu(n) f(n)$,

(b) $\sum_{d|n} \mu(d) f(d) = \prod_{p|n} (1 - f(p))$.

Theorem 9. Let f and g be arithmetical functions.

(a) If f and g are multiplicative, then so is their Dirichlet product $f * g$.

(b) If g and $f * g$ are multiplicative, then so is f.

(c) If f is multiplicative, then so it its Dirichlet inverse f^{-1}.

Theorem 10. For $n \geqslant 1$,

(a) $\varphi^{-1}(n) = \prod_{p|n}(1-p)$,

(b) $\sum_{d|n} \lambda(d) = \begin{cases} 1 & \text{if } n \text{ is a square,} \\ 0 & \text{otherwise,} \end{cases}$,

(c) $\lambda^{-1}(n) = |\mu(n)|$,

(d) $\sigma_a^{-1}(n) = \sum_{d|n} d^a \mu(d) \mu\left(\frac{n}{d}\right)$.

Theorem 11. Let f and g be multiplicative arithmetical functions. Then

(a) $f = g$ if and only if $f_p(x) = g_p(x)$ for all primes p,

(b) $(f * g)_p(x) = f_p(x) g_p(x)$ for all primes p.

Theorem 12. If f is a completely multiplicative arithmetical functions, then

$$f_p(x) = \frac{1}{1 - f(p)x}.$$

Theorem 13. The following are some well-known Bell series.

(a) $\mu_p(x) = 1 - x$,

(b) $\varphi_p(x) = \frac{1-x}{1-px}$,

(c) $I_p(x) = 1$,

(d) $u_p(x) = \frac{1}{1-x}$,

(e) $\lambda_p(x) = \frac{1}{1+x}$.

Theorem 14. Let f and g be arithmetical functions. Then

(a) $(f+g)' = f' + g'$,

(b) $(f * g)' = f' * g + f * g'$,

(c) $(f^{-1})' = -f' * (f * f)^{-1}$.

Theorem 15 (Selberg's Identity)**.** For $n \geqslant 1$,

$$\Lambda(n) \log n + \sum_{d|n} \Lambda(d) \Lambda\left(\frac{n}{d}\right) = \sum_{d|n} \mu(d) \left(\log \frac{n}{d}\right)^2.$$

3 Exercises

Exercise 1. Evaluate

(a) $\mu(n)\mu(n+1)\mu(n+2)\mu(n+3)$,

(b) $\displaystyle\sum_{n=1}^{\infty} \mu(n!)$.

Exercise 2. For each of the following statements either give a proof or exhibit a counterexample:

(a) If $\gcd(m, n) = 1$, then $\gcd(\varphi(m), \varphi(n)) = 1$.

(b) If n is composite, then $\gcd(n, \varphi(n)) > 1$.

(c) If the same primes divide m and n, then $n\varphi(m) = m\varphi(n)$.

(d) If n is composite, then $\varphi(n) \leqslant n - \sqrt{n}$.

(e) There exist infinitely many n such that $\varphi(n+1) < \varphi(n)$.

Exercise 3. Prove that

(a) $\displaystyle\prod_{d|n} d = n^{\frac{\tau(n)}{2}}$,

(b) $\tau(n)$ is odd if and only if n is a square,

(c) $\sigma(n)$ is odd if and only if n is a square or twice a square,

(d) $\displaystyle\sum_{d|n} \tau(d)^3 = \left(\sum_{d|n} \tau(d)\right)^2$.

Exercise 4. Find formulas for the following:

(a) $\displaystyle\sum_{d|n} \mu(d)\varphi(d)$,

(b) $\displaystyle\sum_{d|n} \mu(d)^2\varphi(d)^2$,

(c) $\displaystyle\sum_{d|n} \frac{\mu(d)}{\varphi(d)}$,

(d) $\displaystyle\sum_{d|n} \frac{\mu(d)^2}{\varphi(d)}$,

(e) $\displaystyle\sum_{d|n} \mu(d)^2$,

(f) $\displaystyle\sum_{d|n} \mu(d)\tau\left(\frac{n}{d}\right)$,

(g) $\displaystyle\sum_{d|n} \mu(d)\sigma\left(\frac{n}{d}\right)$,

(h) $\displaystyle\sum_{d|n} \varphi(d)\tau\left(\frac{n}{d}\right)$.

Exercise 5. Prove that

$$\sum_{d|n} \mu(d)(\log d)^m = 0$$

if $m \geqslant 1$ and n has more than m distinct prime factors.

Exercise 6. If $f(n) > 0$ for all n and if $a(n)$ is real with $a(1) \neq 0$, prove that

$$g(n) = \prod_{d|n} f(d)^{a\left(\frac{n}{d}\right)} \quad \text{if and only if} \quad f(n) = \prod_{d|n} g(d)^{b\left(\frac{n}{d}\right)}$$

where $b = a^{-1}$, the Dirichlet inverse of a.

Exercise 7. Let $f(x)$ be defined for all rational x in $0 \leqslant x \leqslant 1$ and let

$$F(n) = \sum_{k=1}^{n} f\left(\frac{k}{n}\right) \quad \text{and} \quad F^*(n) = \sum_{\substack{k=1 \\ \gcd(k,n)=1}}^{n} f\left(\frac{k}{n}\right).$$

Prove that

(a) $F^* = \mu * F$, the Dirichlet product of μ and F,

(b) $\mu(n) = \displaystyle\sum_{\substack{k=1 \\ \gcd(k,n)=1}}^{n} e^{\frac{2\pi i k}{n}}$.

Exercise 8. Let $\varphi_k(n)$ denote the sum of the kth powers of the numbers less than or equal to n and relatively prime to n. Prove that

(a) $\displaystyle\sum_{d|n} \frac{\varphi_k(d)}{d^k} = \frac{1}{n^k} \sum_{m=1}^{n} m^k$,

(b) $\varphi_1(n) = \dfrac{1}{2} n\varphi(n)$,

(c) $\varphi_2(n) = \dfrac{1}{3} n^2 \varphi(n) + \dfrac{1}{6} n \displaystyle\prod_{p|n} (1-p)$,

(d) $\varphi_3(n) = \dfrac{1}{4} n^3 \varphi(n) + \dfrac{1}{4} n^2 \displaystyle\prod_{p|n} (1-p)$,

for $n > 1$.

Exercise 9. Let $P(n)$ be the product of positive integers which are less than or equal to n and relatively prime to n. Prove that

$$P(n) = n^{\varphi(n)} \prod_{d|n} \left(\frac{d!}{d^d}\right)^{\mu\left(\frac{n}{d}\right)}.$$

Exercise 10. Prove the following statement or exhibit a counterexample: If f is multiplicative, then $\displaystyle\prod_{d|n} f(d)$ is multiplicative.

Exercise 11. Assume f is multiplicative. Prove that

(a) $f^{-1}(n) = \mu(n) f(n)$ for every squarefree n.

(b) $f^{-1}(p^2) = f(p)^2 - f(p^2)$ for every prime p.

(c) f is completely multiplicative if, and only if $f^{-1}(p^a) = 0$ for all primes p and all integers $a \geqslant 2$.